Phase 2 Project Submission:

Architecture & Initial Implementation

1. Project Title and Team Information

- **Project Title:**
- **Team Information:** (Names and Roll Numbers)
- **Project Mentor:** (Sir / TA Name)

2. Executive Summary

- Provide a concise overview (1-2 paragraphs) summarizing the project's primary research objective, current development progress, and highlights of Phase 2 achievements. This should serve as a high-level summary understandable without deep cryptographic or technical background.

3. Problem Definition and Scope Refinement

- **Updated Problem Statement:** Refine the exact research gap or technical problem being addressed based on Phase 1 feedback and new insights.
- **Scope of Phase 2:** Detail exactly what was targeted and achieved in this specific phase (e.g., finalized mathematical model, local testnet deployment, baseline benchmarking).

4. Literature Integration & Theoretical Foundation

- **Critical Analysis:** Summarize key research papers and frameworks relevant to the project.
- **Design Justification:** Explicitly highlight how the methodologies from the literature guided your current architectural and cryptographic design choices. Why did you choose your specific approach over existing ones?

5. System Design & Blockchain Architecture

- **System Overview:** Provide a clear, high-level description of the system's architecture.
- **On-Chain vs. Off-Chain Components:** Clearly delineate which parts of your system run on the blockchain (e.g., smart contracts, consensus rules) and which run off-chain (e.g., prover nodes, indexing servers, frontends).
- **Diagrams/Flowcharts:** Include relevant visual architecture. For research projects, this must include data flow diagrams or cryptographic sequence diagrams.
- **Module Breakdown:** Describe the purpose of each major module (e.g., hashing engine, ZK-circuit, RPC node interaction).

6. Mathematical Modeling & Detailed Design

- **Algorithmic/Cryptographic Specification:** Detail the core algorithms, mathematical models, or cryptographic proofs implemented in this phase. Include explanations of the algorithmic complexity or security assumptions.
- **Requirements Analysis:** List the exact software environments, network specifications (e.g., local hardhat node, specific testnet),

and hardware constraints essential for implementing the system.
- **Development Environment:** Outline the specific toolchains used (e.g., Rust, Circom, Foundry, Python) and configuration details.

7. Prototype and Initial Implementation (Proof of Concept)

- **Prototype Description:** Describe the initial working components. For a research project, a "prototype" could be a working Python script of a hash function, a successfully compiled ZK-circuit, or a deployed test smart contract.
- **Repository Link:** Provide a link to the code repository (e.g., GitHub), with clear navigation instructions and a `README.md` explaining how to run the current code.
- **Testing Plans & Baseline Metrics:** Summarize your testing approach. Provide initial performance metrics (e.g., gas costs, proof generation time, hash rate, CPU/Memory usage) and baseline results.

8. Technical Challenges and Solutions

- **Roadblocks:** Identify significant challenges encountered (e.g., compiler limitations, EVM restrictions, mathematical edge cases).
- **Adaptations:** Describe the engineering strategies employed to overcome these challenges and detail any necessary pivots from the Phase 1 proposal.

9. Future Work and Phase 3 Roadmap

- **Phase 3 Objectives:** Outline the exact development and integration tasks planned for the next phase.

- **Remaining Milestones:** Highlight the major hurdles left before the final evaluation and discuss any anticipated resources needed to complete the final IEEE paper.

10. Appendices & References

- **Supporting Documents:** Include data sets, mathematical proofs, extended diagrams, or core code snippets.
- **References:** Cite all research papers, articles, and documentation in strict **IEEE format**.